

M09 Protected Review & Targeted Repair

OPEN ONLY AFTER a genuine saved attempt. Compare reasoning, close Review, then solve a changed fresh task.

DE09-01 - ETL vs ELT and transformation boundaries

Expected reasoning: ETL transforms before final load; ELT lands source-shaped data first and transforms inside the destination. Both need contracts, lineage and reconciliation.

Targeted repair: If you argue only from tool preference, restate the data-risk, recoverability and ownership requirements.

Then close Review and solve a changed variant.

DE09-02 - Raw, staging, clean/intermediate and serving contracts

Expected reasoning: Raw preserves received evidence. Staging standardizes types/keys while surfacing bad rows. Intermediate expresses reusable business logic. Serving exposes stable consumer grain.

Targeted repair: If layers blur together, ask: can I reconstruct source evidence; where is invalid data visible; what table is safe for consumers?

Then close Review and solve a changed variant.

DE09-03 - Deterministic transformations and production SQL style

Expected reasoning: Deterministic work avoids hidden manual edits, unstable ordering and ambient state. Production SQL should make grain, filters, joins and calculations readable.

Targeted repair: If reruns differ, isolate nondeterminism: time-dependent defaults, unordered tie selection, mutable inputs or hidden state.

Then close Review and solve a changed variant.

DE09-04 - Deduplication and key integrity

Expected reasoning: An exact duplicate repeats the same received row. A repeated business key with changed attributes is not automatically a duplicate; choose a version/merge rule and prove it.

Targeted repair: Never add DISTINCT to silence unexpected fan-out. Identify the key/grain first.

Then close Review and solve a changed variant.

DE09-05 - Schema checks and data contracts

Expected reasoning: A contract includes required fields, types, domain rules and evolution expectations. Compatibility decisions should be explicit.

Targeted repair: If a cast fails, preserve the raw value and reason; do not convert every error to zero/null.

Then close Review and solve a changed variant.

DE09-06 - Source-to-target reconciliation

Expected reasoning: Counts alone can pass while money is wrong; sums alone can pass while rows are duplicated. Reconciliation needs both plus exclusion rules.

Targeted repair: If totals disagree, compare layer by layer until the first divergence; repair there.

Then close Review and solve a changed variant.

DE09-07 - Incremental processing and safe reruns

Expected reasoning: Incremental models need a trustworthy cursor and unique key. Reprocessing the same batch must not create duplicates.

Targeted repair: If a changed key duplicates, check unique-key logic and $\geq$ cursor boundary before changing data.

Then close Review and solve a changed variant.

DE09-08 - Failure behavior, retries, quarantine and atomic state

Expected reasoning: State must advance only after the destination commit is durable. Retry transient work; quarantine record-level defects; fail the run for contract-breaking conditions according to policy.

Targeted repair: If state moved on failure, treat it as a correctness bug, not a logging issue.

Then close Review and solve a changed variant.

DE09-09 - Lineage, run metadata and auditability

Expected reasoning: Auditability links source file, run id, layer, counts, state before/after, status and output controls. Lineage records dependencies and grains.

Targeted repair: If you cannot answer which input/run produced a row, the handoff is incomplete.

Then close Review and solve a changed variant.

DE09-10 - Build and defend a reliable batch pipeline

Expected reasoning: A reliable batch path is not just transformation code: contracts -> validation -> quarantine -> deterministic transform -> reconciliation -> tests -> atomic state -> metadata -> rerun proof.

Targeted repair: When a result is plausible but controls fail, trust controls and investigate.

Then close Review and solve a changed variant.

DE09-11 - dbt-style project anatomy, sources and dependency graph

Expected reasoning: source() represents declared external/source relations; ref() creates model dependencies; project/YAML config adds metadata/tests/materialization behavior.

Targeted repair: If the graph is unclear, follow source()/ref() edges before reading every SQL expression.

Then close Review and solve a changed variant.

DE09-12 - Executable data tests, quality severity and CI boundaries

Expected reasoning: Blocking tests protect invariants such as required keys/uniqueness. Warning tests surface risk that may not justify stopping every run. CI should fail only on deliberately blocking conditions.

Targeted repair: Do not make every anomaly fatal and do not make critical integrity checks warnings.

Then close Review and solve a changed variant.

DE09-13 - Incremental models, materialization, documentation and handoff

Expected reasoning: Views, tables and incremental models trade freshness, compute and persistence. Current dbt incremental models require valid SQL for both full/incremental paths; `unique_key` supports updates instead of blind append.

Targeted repair: If incremental logic changes historical meaning, plan a controlled rebuild/backfill rather than assuming future runs repair history.

Then close Review and solve a changed variant.